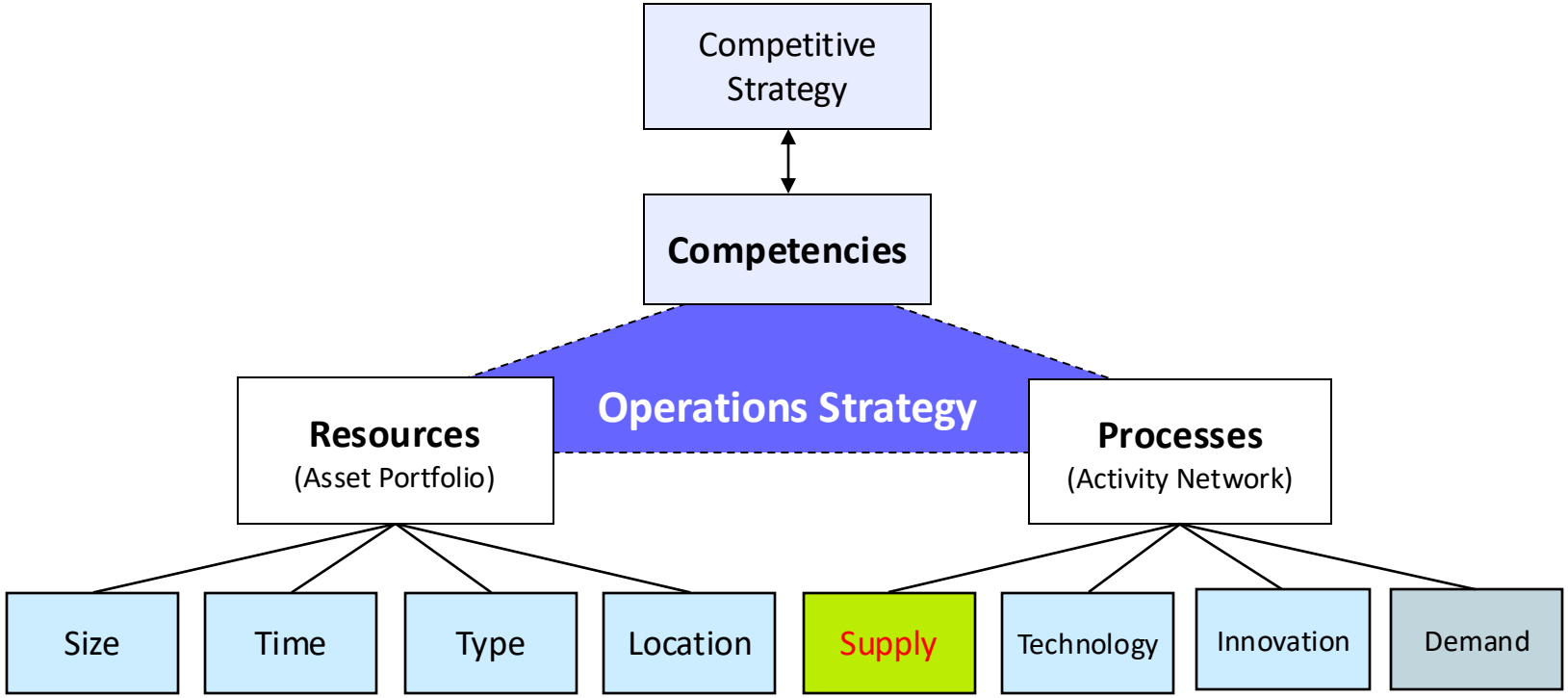
Operations Strategy Framework

In this class, we will discuss how to match supply with demand strategically. Specific topics include strategic sourcing.



Strategic sourcing: Learning objectives

- Be able to
  - Explain the concept of strategic sourcing and integrate it with operations strategy
  - Use a framework to choose a supply relationship and guide outsourcing decisions
  - Articulate
    - TCO
    - Multi-sourcing
    - Tailored sourcing
    - Structured contracts
  - Illustrate how technological change affects sourcing and value chain (dis)integration

Strategic Sourcing: *What is it?*

= Deciding on appropriate supply relationships for each-and-every activity …

- It typically involves:
  - Identify the activity, its requirements and potential suppliers
  - make-or-buy decision (outsourcing, contracting … (dis)integration)
  - Define, establish, and manage the bilateral relationship (CRM, often, but not always, involving contracts)
- Distinguish:
  - Strategic buyers: cross-functional teams (e.g., CPO, global commodity managers) that develop long-term sourcing strategy <u>globally</u> (2 meanings here)
  - Transactional buyers: execute purchase orders, clerical purchasing

Strategic Sourcing: *Why it is important?*

- Sourcing is the biggest single cost for most firms, it accounts for 60% of the average company's total COGS
- Great potential for bottom-line improvement
  - e.g., assume purchasing is 85% of COGS
    - If COGS = $80, then Purchasing costs = .85*$80 = $68
    - Say purchasing costs improves 10% (i.e. $6.8),
    - Then margin improves by $6.8!
- Sourcing must be strategic:
  - Cost containment is fundamental (max NPV)
    - Control costs and prevent wasteful spending
  - If well practiced, it can also drive innovation, quality, flexibility or responsiveness
    - Need talent and mindset beyond transactional purchasing
    - Need to integrate with overall operations strategy

Example: Toyota's CCC21

- Construction of Cost Competitiveness for the 21st Century
  - In the late 1990s, a rival executive said Toyota was paying too much for parts compared to its peers.
  - CEO (then) Mr. Watanabee: "Insulted by the commentary, we launched an internal study that pitted Toyota, component by component, car by car, against other models. The result took months to compile and shattered the myth about the company's cost advantage." In hundreds of components and vehicle systems, Toyota was the winner in just over half. It was "outright humiliating."
  - Mr. Watanabee pushed Toyota and its parts suppliers to tweak the way they designed and made 173 components and systems to make them simpler and less expensive without affecting quality. In one effort, Toyota worked with suppliers to consolidate countless specifications for wire harnesses. That produced tens of millions of dollars in savings.
  - In total, a one-trillion yen ($8.68b) was saved, equivalent to a 30% cut in the company's procurement costs during 2000-2004.

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Integrate Sourcing with Operations Strategy

- Competitive strategy: what is the value proposition to customers and shareholders?
- The value proposition then indicate the priority ranking of competences that sourcing should consider; i.e., what is the relative importance of cost, quality, flexibility, and responsiveness?
- This ranking then guides how the various elements affecting assets and activities should be weighted in making sourcing decisions

Spectrum of sourcing

Market Buy	*Long Term Relationships*	**Vertical Integration**

- Market exchange: external sourcing and transactional
  - Benefits: focus, cost, risk, flexibility, access to latest innovation and technology
- Vertical integration
  - Benefits: control, operational and transactional efficiency, aligned incentives promote optimal investment, learning by doing, protection of proprietary assets, technology and other IP (Intellectual Property)
- Long term relationship parameters
  - Degree of integration
  - Responsibility
  - Comprehensiveness

Strategic Sourcing: a framework

1. Is outsourcing feasible?
   - Is a stable supply base with necessary capabilities available?
   - Is outsourcing politically viable?

No

Yes

Yes

2. Is outsourcing necessary?
   - Are internal financial and operational capabilities insufficient?

No

3. Is outsourcing in line with strategic priorities and risks?
   - Is this activity "non-core"?
   - Is the risk of outsourcing it tolerable?

alignment

No

Yes

4. Is outsourcing desirable given our value proposition?
   - Can external suppliers do it better? (**TCO** & NPV)

No

Yes

5. Do we have the ability to manage suppliers and ongoing risk?
   - Can we **contract** on detailed requirements?
   - Can we coordinate incentives and operational flows?

tailor

Easy

Difficult

Impossible

Market Buy | *Long Term Relationships* | **Vertical Integration**

Strategic sourcing: TCO

- What are the costs associated with a supplier?
  - TCO evaluates the total cost of sourcing or of using any activity provided by a given supplier
  - TCO is calculated following 3 steps:
    1. Determine all activities impacted by (the particular) sourcing
    2. Identify and quantify cost drivers using activity-based costing
    3. Calculate TCO of each potential supplier
  - TCO uses
    1. Support make-or-buy decision and supplier selection
    2. Communicate TCO to manage suppliers over time
    3. Communicate TCO to customers (to sell or to coordinate)
    4. Improve coordination and performance across supply chain

Strategic Sourcing: multi-sourcing

Arguments in Favor of Single Sourcing

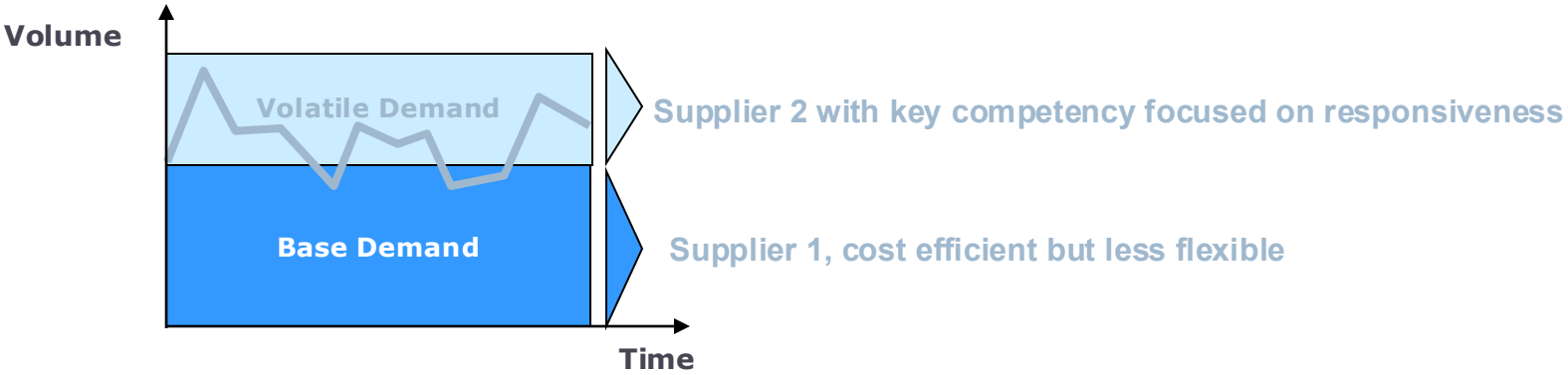
- The supplier may be only available source
- Quantity is too small to split
- Opportunities for discounts or lower freight costs
- Outstanding quality, service, value
- More important customer more attention from supplier
- Focus on one, not many suppliers
- Prerequisite to partnering
- Patent protection, exclusive distributorship
- Easier to schedule deliveries
  - JIT, stockless buying and arrangements
- …

Arguments in Favor of Multiple Sourcing

- Supply market volatility
- Limited supplier capacity
- Assurance of supply
- Avoid supplier dependence
- Obtain a greater degree of volume flexibility
- Opportunity to test a new supplier
- Strategic considerations; e.g., military preparedness
- Government regulations
- Keep suppliers “on their toes”
- …

Strategic Sourcing: tailored-sourcing

- Possible dimensions along which sourcing can be tailored
  1. Volume and responsiveness-based
  2. Product or customization-based
  3. Quality-based
  4. Innovation & technology-based



Strategic Sourcing: structured contracts

- Quantity Discount Contracts
  - All-unit quantity discount
  - Incremental quantity discount
  - Cumulative/volume discount
- Quantity Flexibility Contracts
  - committing a certain % of forecasted quantity before demand is known so suppliers are willing to invest in extra capacity to anticipate possible higher demand
- Buy-back contract
- Revenue Sharing Contracts
- Options Contracts
- …

Technology affects sourcing and value chain

- Technology change affected sourcing and value chain (dis)integration
  - Electronics industry
  - Automobile industry
  - Service industry
- Outsourcing is more attractive when
  - Product technology is modular with simple interfaces and modules can be standardized across industry
  - Process technology can be separated from product technology and is highly specialized, or benefits from strong economies of scale, scope, or risk pooling
  - Transportation and information technology: transportation of goods and transmission of information is relatively inexpensive compared to the product or service value